

Zitong Tian

 [Homepage](#)

EDUCATION

2023 – 2028	PhD (Mathematics) at Qiuzhen College, Tsinghua University (GPA: 3.99/4.0) Advisor: Prof. Juncai He
2023.02 – 2023.06	Enhanced Program for Graduate Study at Beijing International Center for Mathematical Research, Peking University
2019 – 2023	Bachelor's Degree in Mathematics at School of Mathematics Science, Shanghai Jiao Tong University (GPA: 4.05/4.3, Rank: 1/18)

RESEARCH EXPERIENCE

Scientific Computing and Machine Learning [Math] May 2025 – Present
Propose an explainable framework to solve a class of PDE with conservation law with high accuracy. The paper is submitting.

Develop L^p space theory and approximation of shallow neural networks via micro local The paper is submitting.

Molecule and Peptide Generation [AI for Science] Dec 2024 – Feb 2026
Develop a fixed-length autoencoder for molecular structures to enable molecular editing and optimization. This work is accepted to NeurIPS 2025.

Propose Axial Feature Injection for chirality peptide design with interesting theoretical insights. This work is accepted to ICML 2026.

INTERNSHIP

Research Intern at [Sapient Intelligence](#) Apr 2026 – Present

- Research on the HRM model for PDE solver, together with PDE foundation models and new neural operators.
- Research on new generative models for images.
- Develop a Agent system for Alpha mining from scratch.

Research Intern of AI for Science at [AIR](#) Dec 2024 – Feb 2026
Research on the generation of molecules and peptides.

TEACHING

Teaching Assistant, Machine Learning and Differential Equations	Sep 2025 – Jan 2026
Teaching Assistant, Probability II	Sep 2024 – Jan 2025
Teaching Assistant, Analysis II	Feb 2024 – Jun 2024
Teaching Assistant, Ordinary Differential Equations	Sep 2023 – Jan 2024

PUBLICATIONS

($\alpha - \beta$) denotes alphabetical authorship; * denotes equal contribution.

Juncai He and **Zitong Tian** ($\alpha - \beta$) (2026). *Sharp Sobolev Sandwich and Approximation Rates of Radon-Domain L^p Ridge Integral Spaces for ReLU^k Networks*. arXiv: [2606.24795 \[math.NA\]](#).

Juncai He, Xinliang Liu, and **Zitong Tian** ($\alpha - \beta$) (2026). *Divergence-Free Linearized Neural Networks: Integral Representation and Optimal Approximation Rates*. arXiv: [2603.28638 \[math\]](#).

Ziyi Yang*, **Zitong Tian***, Yinjun Jia*, Tianyi Zhang, Jiqing Zheng, Hao Wang, Yubu Su, Juncai He, Lei Liu, and Yanyan Lan (2026). “Cross-Chirality Generalization by Axial Vectors for Hetero-Chiral Protein-Peptide Interaction Design”. In: *ICML, 2026*. URL: <https://arxiv.org/abs/2602.20176>.

Zitao Chen*, Yinjun Jia*, **Zitong Tian***, Wei-Ying Ma, and Yanyan Lan (2025). “Manipulating 3D Molecules in a Fixed-Dimensional E(3)-Equivariant Latent Space”. In: *NeurIPS, 2025*. URL: <https://arxiv.org/abs/2506.00771>.

AWARDS

Ph.D. Period

Outstanding Student Leader of Tsinghua University	2026
The Comprehensive First-Class Scholarship of Tsinghua University	2025
Outstanding Communist Youth League Member	2024

Undergraduate Period

Outstanding Graduate of Shanghai Municipality	2023
Academic Excellence Scholarship, Dong Scholarship, Liu Yongling Scholarship	2020-2023
Outstanding Communist Youth League Member, “Three Goods” Student	2020-2022
Men’s 400m, University Track and Field Games	6th Place
Men’s 1500m, University Track and Field Games	9th Place

Middle School Period

China National Mathematics Olympiad	First Prize
China Southeast Mathematical Olympiad	Golden Medal
President’s Cup Football Tournament	Champion
Men’s 800m, Track and Field Games	3rd Place
Provincial-Level “Three Goods” Student	

SKILLS

Solid math and machine learning theory background, Python Coding skills.